

# REPORT

## Title

### Prolog Program for Medical Diagnosis

## Aim

To implement a **simple medical diagnosis system using Prolog** that identifies a possible disease based on given symptoms.

## Problem Statement

Develop a Prolog program that diagnoses a possible illness by checking the symptoms provided. The program should use logical rules to match symptoms with diseases such as flu, common cold, malaria, and typhoid.

## Requirements

- SWI-Prolog
- Computer system
- Prolog source file (`exp12.pl`)

## Procedure

1. Open **SWI-Prolog**.
2. Create a new file named `exp12.pl`.
3. Define symptoms and diagnosis rules.
4. Save the program.
5. Load the program using `[exp12]..`
6. Enter a symptom-based query.
7. Observe the possible diagnosis.
8. Use `;` to check for other possible diagnoses.

## Program

```
% Medical Diagnosis System

diagnose(flu) :-
    fever,
    cough,
    body_pain,
```

```
fatigue.

diagnose(common_cold) :-
    cough,
    sneezing,
    runny_nose.

diagnose(malaria) :-
    fever,
    chills,
    sweating,
    headache.

diagnose(typhoid) :-
    fever,
    headache,
    stomach_pain,
    weakness.
```

```
% Symptoms
fever.
cough.
body_pain.
fatigue.
sneezing.
runny_nose.
chills.
sweating.
headache.
stomach_pain.
weakness.
```

## Sample Output

### Query:

```
?- [exp12].
true.
```

```
?- diagnose(flu).
true.
```

### Another query:

```
?- diagnose(common_cold).
true.
```

To find all possible diagnoses:

```
?- diagnose(Disease).
```

### Output:

```
Disease = flu ;  
Disease = common_cold ;  
Disease = malaria ;  
Disease = typhoid.
```

### Conclusion

Thus, a **simple medical diagnosis system was successfully implemented in Prolog** using facts and rules. The program identifies possible diseases by matching the available symptoms with predefined diagnostic rules.

**Note:** This is an educational Prolog example only and is not a substitute for professional medical diagnosis.